BANKHAUL · DESIGN DECISION RECORD

BankHaul requirement decisions, round 2

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DOCUMENT CONTROL

Document ID	BKH-DDR-003	Version	0.2
Status	Superseded	Date	2026-10-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/bankhaul		
Source	docs/decisions/0003-requirement-decisions-round2.md @ 2177c86		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-25	Amish Chadha	Superseded by BKH-DDR-003 in 0003-amish-requirement-decisions.md, which records the same two decisions (BKH-DEC-001 O-5, decided by Amish 2026-10-25)
0.1	2026-10-03	Amish Chadha	R5 (far stake in very soft mud) and R8 (cost per station) decided by Amish on 2026-10-03, both as recommended

0003: Requirement decisions, round 2

> **Superseded** (Amish, 2026-10-25, BKH-DEC-001 O-5): this record duplicates the ID BKH-DDR-003 and the two decisions of `0003-amish-requirement-decisions.md`, which is the record kept. It stays here for history only.

- **Date:** 2026-10-03
- **Status:** decided. Decided by Amish Chadha, 2026-10-03: "i approve all of the 47 recommendations provided by you. Execute them." Both decisions below were the recommended options in `docs/REVIEW.md` (TRL 3, Decisions for Amish) and in the design decisions register, taken exactly as worded there.

Context

At TRL 3 (BKH-CAL-001 v0.1) two requirements were not met or at risk on paper. R5 (far stake holding) was met in soft mud but at risk in very soft mud: in a bed of about 5 kPa the 76.1 mm stake driven 2.5 m holds an ultimate 1,845 N, a factor of 1.23 on the 1.5 kN load, against the factor of 2 used as "holds", because lateral capacity in clay falls with its strength and the load acts 1.2 m above the bed. R8 (parts under USD 80 per station) was not met: the constructable station costs USD 382, driven by bought rated hardware and the steel stake and post.

> **Safety:** Neither decision relaxes a safety provision. The long stake is heavier (33.2 kg) and is handled by two people from a boat held on two anchors, with a lookout, as before; the hand vane reading of the bed is taken under the same rules. The steel trial station keeps the rated rigging that carries the safety case.

Options considered

Table 1. Options for each decision.

Requirement	A	B	C
R5, far stake holding in very soft mud	Drive 3.5 m into very soft beds (5.0 m stake): factor 2.03, met, about USD 12 more per stake and 6.4 kg more (33 kg stake)	Two stakes 1 m apart tied at the top with a bar carrying the collar: factor 2.21, met, about USD 56 more and about 28 kg more, two to drive from the boat	Keep 2.5 m and site stations only where a hand vane shows 10 kPa or more: factor 2.46 where used, very soft beds excluded, no cost or mass
R8, cost per station	Build this steel station for the TRL 4 trials and cost a local-materials station with the co-design partner before the season pilot: R8 not met for the trial station, USD 382 now, about 100 kg	Switch the prototype now to local materials: about USD 98, not met but close, about 30 kg more, R7 and R9 at risk	One bank post and table serve three loops at a landing site: about USD 257 per loop, not met, about 60 kg per loop

Decision

Table 2. Decisions taken.

#	Requirement	Option chosen	Effect	Condition
5	R5, far stake holding in very soft mud	A: drive 3.5 m into very soft beds with a 5.0 m stake	Factor 2.03 at 5 kPa (ultimate 3,040 N), 2.46 at 10 kPa with the standard 2.5 m: R5 met on paper. Long stake 33.2 kg (26.8 kg standard), about USD 12 more; station USD 394 at a very soft site. Bending stress unchanged (162 MPa, factor 2.2 on S355)	The extra 1 m is cut only where a hand vane test of the bed, from the boat down to 2.5 m, reads under 10 kPa. It gives a factor of 2 only down to 4.9 kPa; weaker beds are a new open question
6	R8, cost per station	A: build the steel station for the TRL 4 trials; cost a local-materials station with the co-design partner before the season pilot	R8 not met for the trial station (USD 382, or USD 394 with the long stake; about 100 kg). No change to the design or the bill of materials	The local-materials costing, from local prices with the co-design partner, is made before the season pilot and gives the real R8 figure

Consequences

- `cad/src/model.py` : `embed_vsoft` (3,500 mm) and `su_vane_min` (10 kPa) added, with the derived length of the long stake (5,000 mm). The model and the STEP files still show the 2.5 m design case, so they were not re-exported.
- `cad/src/sheets.py` : far end general arrangement BKH-DWG-002 to Rev P3 with the long stake note; `cad/src/build_plan_media.py` : making sketch BKH-DWG-102 with the long stake's cut length and paint ring.
- `docs/04-calcs/sizing.py` and `results.csv` (BKH-CAL-001 v0.2): R5 met on paper; the weakest bed covered (4.9 kPa); station cost and heaviest lift for the long stake; R8 restated as not met for the trial station.
- `bom/bom.csv` : line 2 note gives the long stake's cut length, mass and extra cost; the costed station stays the standard one (USD 382).
- Build plan BKH-BLD-001 v0.2: hand vane reading of the bed before the stake is cut, the long stake's cut length and paint ring, and the vane reading added to the safety stop before driving.
- `docs/03-requirements.md` v0.3: R5 and R8 status updated; no target restated.